dPCR PROTOCOL – 96 well plate

Quick Overview

Parameter	Information
Protocol ID	MOL004
Version	1
Method Owner	Lieke Lipsch
Estimated Hands-on Time	2.5 hours
Total Duration	5.5 hours
Difficulty Level	☐ Basic ☒ **Intermediate** ☐ Advanced
Training Required	☒ **Yes** ☐ No
Biosafety Level	/
Typical Throughput	94 samples (up to 4*94 samples per day)
Sample types	Extracted DNA
Output	Copy number of a gene/ uL
Related Protocols	MOL001 – DNA extraction MOL005 – qPCR/dPCR Assay details for ARGs/intI1/16S rRNA

Method Information

Method Title: dPCR protocol

Author(s): Lieke Lipsch

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Last Updated:16-06-2026

Reviewed By: Quinn Hinkel, Franziska Raab

Approved By:

Purpose

This method is used to measure the number of copies of certain genes within samples.

Scope

Suitable for

- Extracted DNA

Not suitable for

- Extracted DNA with very low quality (Nanodrop values)

Safety Considerations

Hazards

- Biological: /
- Chemical: /
- Physical: /

Required PPE

- Lab coat
- Gloves

Waste Disposal

- Biological waste: tips – bag to autoclave, plates – place in plate waste bag
- Chemical waste: /
- Sharps: /

Materials and Reagents

Reagent/Material	Specification	Supplier	Storage
Probe PCR Master Mix + RNase free water	QIAcuity Probe PCR Kit (ID. 250102)	Qiagen: https://www.qiagen.com/us/products/instruments-and-automation/accessories/qiacuity-probe-pcr-kit?catno=250102	-20 degrees freezer
96 well dPCR plates	QIAcuity Nanoplate 8.5k 96-well (ID. 250021)	Qiagen: https://www.qiagen.com/us/products/instruments-and-automation/accessories/qiacuity-nanoplates-and-accessories?catno=250021	Room temperature – PCR lab
Probes Primers	synthesised on demand, assay specific (**More information MOL005**)	Biomers: https://www.biomers.net/	-20 degrees freezer
BSA	Concentration of 2.5 mg/mL (**only needed for phenol-chloroform extracted samples instead of kit**)	E.g. Thermofisher: https://www.thermofisher.com/order/catalog/product/de/de/AM2616	-20 degrees freezer

Equipment

Equipment	Model	Location
QIAcuity Digital PCR System (dPCR) UV bench	Qiacuity Four Platform System (ID. 911042)	Lab 105 PCR lab

Procedure

Before Starting

Checklist

☐ Required lab training completed

☐ Required eplas training completed

☐ Equipment available/ reserved

☐ Sample IDs assigned

1.Before starting in the lab

Put your sample labels in the csv plate layout template, it is easiest to only change the sample names and adapt the rest in the dPCR software.

Template can be found here:
https://drive.google.com/file/d/1OkCAZvAU1gwobCJHpbfCX4KBxyMZYobf/view?usp=sharing

2.Preparations dPCR

1. Check which dilutions to use for your samples to get the best results
2. Prepare primers and probes

2.1 Test which dilution works best for your samples

This only needs to be done once at the start of a new project and does not need to be repeated for each dPCR. When you have found the dilution that work for your samples you can continue using these.

Use dilutions of your samples:
Genes: original sample, 1:10, 1:100
16S: 1:100, 1:500, 1:1000, 1:10.000, 1:100.000

10 uL of your original sample can be put into the first tube of a PCR strip.
1:10 dilution: 1 uL sample + 9 uL PCR clean H2O
1:100 dilution: 1 uL 1:10 dilution + 9 uL PCR clean H2O
1:500 dilution: 2 uL 1:100 dilution +8 uL PCR clean H2O
Etc. (Alternatively you can use 2 uL of sample and double the amount of PCR clean H2O).

It is best to make fresh dilutions on the same day to ensure correct results (especially dilutions > 1:1000). Vortex very well (10 sec) in before pipetting to the next dilution. Use 0.2 ml strips.
Hint: For kit extracted samples 1:1000 was used for 16S (and manure samples) and 1(undiluted) sample for all other genes. For phenol-chloroform extractions : dilution 1:1000 for 16S rRNA and manure samples and dilution 1:10 combined with BSA for intI1 and the other resistance genes.

2.2. Prepare primers and probes

All of our diluted primers have a concentration of 10 uM, the undiluted primer stock has a concentration of 100uM and will be labeled as such. If you have to dilute from the stock follow this example:
Concentration primer 100 uM
1:10 dilution: 1 uL 100uM primer/probe + 9uL PCR clean H2O or alternatively:
2 uL 100uM primer/probe + 18uL PCR clean H2O

3.Prepare MasterMix (volumes for 8.5k 96 well plates)

Total volume per well is 12 uL, this includes 2 uL of DNA template (so you need 10 uL of master mix per well). Times this to the number of wells you are planning to fill including some extras and at least one negative. For each gene you are testing, you need to make a separate mastermix. Make sure that for the total volume you have round numbers or a pipette that can pipette the right volumes.

When using phenol-chloroform extracted samples, BSA is added to the mastermix, and less RNase free water is added.

When you are pipetting a full plate, you can pipette your mastermix into a PCR strip to make it easier to use the multichannel pipette.

	Volume per well (uL)	Volume for a whole plate (uL):	Total volume (incl. some extras and at least one negative)
Probe PCR Master Mix	3		
Primer - F	0.96		
Primer - R	0.96		
Probe	0.48		
RNase free water	4.6/**3.6**		
DNA (option: diluted)	2		
BSA (2.5 mg/mL)	**1**		

4.Prepare plate

1. Decontaminate UV bench for at least 10 minutes with air circulation and H2O2.
2. Thaw completely, vortex, and spin down everything you will use. Note: probe is not light stable, keep it in the dark. Put everything in fridge or cooling block when you're not using it.
3. Take a 96 PCR clean plate, only open in UV bench.
4. Pipette your Mastermix into PCR strip for multichannel pipetting.
5. Take a dPCR plate, cover film, and plate holder. Don't touch the underside of the plate. Put the plate in plate holder and cover the plate by laying the film on top of the plate.
6. Pipette in 10 uL of mastermix into each well of the 96 PCR clean plate.
7. Then pipette your DNA into the mastermix, per column.
    - Pipette up and down (5x), and pipette 10 uL into the dPCR plate without any bubbles. Do not press until second pressure point of the pipette. Pipette slowly, at a 45 degree angle, into the plate but without touching the bottom of the plate.
    - Include at least one blank per gene with only mastermix.
8. Stick on foil:
    - Pull off the white part of the cover film, lay the blue part in the middle of the plate
    - Use the roller from dPCR drawers to secure it.

- Tighten with your fingers. Take off clear foil while holding down the blue film with your hands
- Use a lot of pressure to roll down the blue film, until you can see the structure of the plate, also roll on edges. Don't move the plate while doing this.

General:

- Mastermix can be stored in the fridge for a few weeks but is better stored in freezer when you are not using it often.

5.dPCR machine

1. Start up the laptop next to the dPCR machine. Password laptop = Admin
2. Open the Qiacuity software suite (on desktop). You will get a warming message that the connection is not private. Go to advanced > proceed to local host (unsafe)
3. You can login to the software and the machine using login = admin, and password = admin.
4. Create a new plate for each experiment (can be made into a template too, to reuse).
5. Go through steps for the plate
    - Name plate: include your name, gene names, sample names (optional dilutions)
    - Select 8.5K, 96 well plate
    - Description/ labels/ plate ownership can be skipped
    - Make sure you use the right plates! 96 well plates are black and gray (not blue!)
    - Scan barcode of plate (on side). Find barcode scanner in drawers below dPCR machine and connect via usb to the laptop. Scan side of plate until you hear a beep. Remember which plate you scanned for which digital plate if you are working with more than one plate (or number your physical plates).
    - Put scanner away.
    - Safe changes.
1. **dPCR parameters** →priming probe (not eva green) → This part cannot be changed after the run!
    - Cycling (at fixed temperature):

Step	Time	Temperature (deg C)	Cycles
PCR initial heat activation		95	1x
Denaturation	15s	95	40 cycles
Annealing	15s 15s	52 72	

    - Safe changes
7. Reaction mixes:
    - Create different reaction mixes when you use multiple mastermixes for multiple genes on one plate.
    - Total volume: 12 uL (per well), add for each

Target name:	Dye: (can be found on primer tube)	Channels:
16S rRNA	FAM	Green
sul1	Hex	Yellow
sul2	FAM	Green
intI1	Hex	Yellow
tetW	Cy5	Crimson
tetA	Cy5	Crimson
ermB	Texas red	Red

8. Imaging→ most important of all:

- You can turn off the colours that are not shown up above in the channel. This reduces stress on the samples.
- You can use 2 imaging moments if you are unsure about the amount of cycles that work for you.

8. Samples and control:
    - You can add this later
8. Plate layout:
    - Can be added later
    - Select what you want to define and name the wells, which sample is in which well.
        - This can be done by importing the csv file made in the 1st step.
    - Include template volume (amount of DNA you add): 2uL
    - Can include concentration factor here too in case you diluted your sample.
    - Save changes and click done button.

General dPCR machine:

- You can prepare the program in advance.
- If you add more imaging steps the colours are selected correctly automatically.
- More steps can be added after running your plate.
- You can abort a run by using a stop button (best before priming).

6.Load plate

1. Log into the machine.
2. Open the tray by pressing the button.
3. Put in plate, with barcode facing the back of the machine.
4. You can see that the plate is loaded and you can press play/ run all on the program you made.

7.Analyses

1. Check source image, repeat samples that only have half filled wells.
2. Check the 1D plot, the thresholding can be done manually if necessary.
3. Results can be exported to csv files (page 211), and loaded into Excel (page 215), column 14 shows the concentration that can be used for further calculations (Conc. [unit] (undiluted sample)) see QIAcuity User Manual (on desktop dPCR computer). https://drive.google.com/file/d/1fzIch29fi7EdsXxJ16_-_Ymfkucrtlte/view?usp=sharing

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Critical Control Points

⚠ Step: Pipetting without bubbles

Why it is critical: It will make it impossible for the machine to analyse your samples when there are air bubbles.

How to avoid mistakes: For the first time do this together with someone who has done it before. Practice the pipetting before starting and work carefully/ slowly.

Quality Control

QC Parameter	Acceptance Criterion
Number of filled nano partitions in the source image	More than half of the 8500 nano partitions are visible on the source image

Troubleshooting

Problem	Possible Cause	Solution
Inhibition	DNA extraction method	Use 1 µL BSA (2.5 mg/ml) and reduce amount of PCR water by 1 µL

Expected Outcome

A successful result look like a fully filled cell and shows a realistic gene copy number per uL.

Data Recording in Lab Books

Record the following information:

- Operator
- Date
- Sample IDs
- Batch numbers
- Deviations from protocol
- Raw data location

Storage and Stability

Item	Storage Condition	Stability

Plates	Fridge	Can be stored for a few hours

References

- SOPs
- Publications
- Instrument manuals: https://www.qiagen.com/us/resources/usermanual/qiacuity-um-v3-5

Revision History

Version	Date	Changes	Author
1.0		Initial version	
	11.9.26	Added comments, missing information and corrected numbering mistakes	Quinn Hinkel
	05.10.26	Added comments, missing information and some questions	Franziska Raab